Potential **RCT** and decarbonization course topics

This document will outline suggested projects for the 2026 spring semester edition of the Decarbonization Data Lab. There are two main types of projects from which students can choose:

1. **Research Design:** Students will use the suggested topic to outline a problem and suggest the relevant hypothesis and intervention to address that problem. The main output will be a pre-analysis plan, a document typically produced when preparing to begin an experiment in multiple disciplines, such as development economics. This document must outline key components of the experiment that should be known beforehand, such as logistics for gathering units for the study, providing justification for the number of units to gather using power analysis informed by pre-existing literature, and outlining the causal structure present and addressing it appropriately using techniques from experimental design.
2. **Literature Review:** Students will comprehensively survey a subject to understand the present state of scientific knowledge in a given area. The main aim is to produce a literature review, formatted similarly to those published widely in academic journals, which highlights where current research on a problem is and is not focused. Students should try to identify research gaps with clear potential for experimentation. Students could also provide critiques of the present state of the literature, if it is deemed fit, following a discussion with the teaching team.

The list below is of suggested topics for students. Most topics below could be applied to either format. Students should select which project format they aim to pursue once they have selected a topic.

The list below is NOT an exhaustive list of all topics on which students can choose to work. Students can come to the teaching team with their own ideas or different takes on the subjects listed below. The focus of any proposed ideas should broadly fit within the decarbonization theme of the course.

1. **Improving the SPARK laboratory sustainability program using the BEATRIX behavioural AI tool from FehrAdvice:** sustainability in natural science laboratories is a priority for any university due to their large material throughput and resulting environmental impact. The SPARK Program is a list of targeted measures to improve natural science laboratory operations. It is a novel program in its pilot phase and, therefore, particularly open to further refinements using behavioural economics insights. Therefore, various students or groups of students can analyze the list of actions using the BEATRIX tool, to which course students have exclusive access for the duration of the course, suggest improvements and design experiments to test their implementation. This topic has the potential to create several smaller RCTs.

2. **Ultra-Low Temperature (ULT) freezer change challenge**: changing the temperature of ultra-low freezers from -80 to -70 degrees is a particularly impactful single measure. Researchers unwilling to change the temperature of the ULTs provide various reasons, such as security provided by the lower temperature in case of an electricity outage. Starting with a survey of these potential reasons, various information treatments such as calls for actions by senior scientists for changing the ULT temperatures, social norms by modelling of faculty leaders or studies of safe storage temperatures can rigorously be tested in various field experiments. Reviewing the literature in this field, which might be populated already, is a necessary first step to add value to the current debate with your project.

3. **Extending the Marginal Abatement Cost Curve Approach to a Marginal Value of Public Funds (MVPF) Approach**
   Building on the existing marginal abatement cost curve (MACC) methodology, this project extends the MACC methodology with a framework that incorporates the marginal value of public funds. The MVPF approach can be used to evaluate the cost-effectiveness of policies identified in the UZH core strategy for climate neutrality by comparing the benefits of emission reduction measures to their costs, adjusted for the value of public funds. This extension would involve integrating economic and environmental data to create a comprehensive model that assesses the trade-offs between different policy options. The research could provide a more nuanced understanding of how to allocate public resources efficiently to achieve climate goals. This can be applied to existing and planned decarbonization measures at UZH (more details to be provided once approved). Testing the MVPF with an RCT, for example, the willingness to pay for a certain measure, would make the analysis well-rounded.

4. **Green public procurement**: Procuring greener products in terms of lower emissions can substantially reduce the scope 3 emissions of universities. Similar to a carbon price, it can create a predictable source of demand for novel

products, thus creating a potential lever for green innovation. As awareness of this potential lever is currently low, can treatment with an information video increase low-carbon procurement in "treated UZH units" vis-à-vis a control group? Proposing other measures and potential field experiments is also encouraged, for example, in the UZH procurement platform "P4U", or the SPARK platform, which includes a calculator for lab equipment and will shortly be rolled out at UZH.

5. **Leveraging the Living-Lab/Reallabor-Approach for causal evidence on campus decarbonization**
   The concept of universities as "living labs" and the Reallabor-Approach (Parodi and Steglich, 2021) are gaining traction at university campuses like UZH and ETH. This literature review lists past interventions on university campuses that allowed causally inferring treatment effects of the various interventions and highlights novel areas for conducting RCTs on university campuses such as UZH and ETH. Examples include POWBal trials in LSE students' halls, randomizing carbon prices at Yale, nudging sustainable cafeteria consumption at LSE and the University of Bonn, for example. The literature review should structure RCTs and other quasi-experimental methods in a convincing methodological framework that also clearly identifies gaps and potential opportunities for impactful trials in the future. This would require the students to familiarize themselves with the Reallabor and living lab concept independently, as it is not part of the course curriculum per se. This would primarily be a literature review topic.
   Citations: Parodi, O. and Steglich, A., 2021. Reallabor (pp. 255-265).

6. **Low carbon procurement of lab supplies**: Lab supplies constitute a major source of scope 3 emissions for the UZH. Novel AI-informed approaches allow not merely for the creation of product-sharp carbon footprints, but also for the tailoring of advice for individual suppliers and their individual products, such as changing the source of energy generation, input materials, or shipping options. Leveraging the insights from the Terralytiq platform, student researchers actively engage with suppliers on their decarbonization levers, measure progress, and potentially inform UZH students and staff about their role in further reducing the UZH procurement spent footprint via randomized interventions.

7. **Shifting Consumer Preferences from High to Low-carbon goods – an economic complexity approach:** We have created so-called product carbon footprints for various procured products of UZH with the help of Terralytiq, yet it is hard to identify substitutes in a dataset of 3600 products. The economic complexity approach that Andres et al. 2023 have used to map clean and dirty products and their respective distances in the product space might help to find approximate answers to this question.

8. **Causal Evidence on What Works for Energy, Decarbonization and Climate Policies (Literature Review)** According to a recent paper in Science, empirical evidence on "what works" in climate policies is scarce (Stechemesser et al., 2024). Therefore, the students will conduct an evidence review on "what works" in climate policies, for example, in carbon dioxide removal policies, actively engaging in the What Works Community. Potentially, they could present it at the What Works Summit in Berlin 2026, for which they should apply for the ULF Engage Now funding Call. There is also the option for students to conduct a replication or reproduction analysis of the studies on evidence-based policy making.

9. **RCT on German Firm Tax Credit Use:** In 2020, Germany introduced the "Forschungszulage" (a R&D tax credit) to expand participation in innovation by reducing the effective cost of eligible research activities. The program provides firms with a tax allowance of up to €2.5 million per year, equivalent to 25% of a maximum tax base of €10 million (Federal Ministry of Finance, 2019). Small and medium-sized enterprises (SMEs) are eligible for an enhanced rate of 35%, translating into potentially higher tax relief for the same level of R&D expenditure. Despite these incentives, adoption remains uneven. SMEs—particularly those in clean-technology sectors such as heat pumps, hydrogen, and low-carbon materials—report low program awareness, unclear eligibility criteria, uncertain returns, and perceived administrative burdens. This informational and procedural complexity helps explain why SME application rates lag those of larger firms and why a neutral, scale-invariant policy instrument has yet to reach its intended audience effectively.

   German is required for this project. Potential interviews in Germany are possible and strongly encouraged, for which students should apply for the ULF Engage Now funding Call. The interviews would be to ascertain the reasons firms give for why they are not applying for funding.

10. **Pushback on Environmental reforms – a literature review of commonly cited reasons for resisting the implementation of sustainability measures:** The transition to sustainable practices is a place of both agreement and resistance. As this does not just come from actors protecting their own financial interests (i.e. fossil fuel companies), it is important to understand why people and organizations can be resistant to sustainability measures. In the university setting, this takes the form of anything from resisting reforms to change flying habits to not adjusting purchases to reflect sustainable production. Though literature exists on other forms of sustainability discord (Patterson et al. 2024), a robust examination of pushbacks in the university setting does not appear to

exist. Students will review papers on individual programs, such as the ULT freezer temperature interventions mentioned above, to catalogue common pushbacks and find themes that could lead to an intervention to help mitigate pushbacks to decarbonization measures.

11. **Review of AI-generated literature on energy policy subjects**: As LLM use begins to increase policy development, students should practice assessing the validity of important recommendations made by common LLM models. In this **literature review**, students will review the literature on an energy policy subject of their choice and assess how well a common LLM makes recommendations when prompted to create a policy brief on the same subject. They should highlight easy-to-find issues (i.e. fake citations) as well as in-depth issues, such as factual inaccuracies, contradictory recommendations, and vague descriptions (among others). By comparing this document to their reviewed literature, they will be able to improve on those recommendations and draft a warning document on what sorts of issues consistently appear in LLM-generated policy briefs.

12. **Cafeteria Improvement at UZH:** The UZH has proposed multiple initiatives to improve the sustainability of food consumption on campus. Measures such as No Butcher Mondays and the continued use of the carbon footprint classification system (for more information, see here). There remain multiple avenues for progress. Students are encouraged to consider questions about new methods and how they can complement existing programs or examine how already-deployed methods may be missing their full potential.

13. **Incentivization of energy efficiency in buildings:** The weatherization study conducted in Michigan by Fowley et al. (2015) was a case study to assess how successful home refurbishment incentivization schemes have been in their actual energy and cost savings for participants. This particular program, one of the largest in the United States, was found to be severely underperforming its forecasts, and the additional interventions to improve uptake still proved very costly. Students are invited to examine other refurbishment, insulation, or retrofit projects around Switzerland and Europe to see how these programs differ from the Michigan program. This allows students to use the weatherization study as a template whilst taking into account the experimental and logistical realities of the new location, along with considering how the results of such a study generalize to the non-US context and providing important comparative analysis.